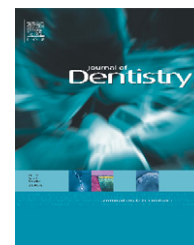


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Enamel—A “metallic-like” deformable biocomposite

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ABSTRACT

Objectives: The aim of this study is to compare the mechanical responses of enamel with dental-used metals and to show that enamel has mechanical properties similar to metals rather than ceramics.

Methods: Four dental-used metals: cast alloy, gold alloy, titanium and amalgam, were compared with enamel. Pure hydroxyapatite (HAP) was used as the ceramic analogue of enamel. A Berkovich and a spherical indenter were selected for nanoindentation tests. The stress-strain (H - a/R) curves and indentation creep behaviour were investigated and analyzed.

Results: Although the primary composition of enamel is hydroxyapatite, the indentation stress-strain curves and creep behaviour of enamel was totally different to HAP. Enamel had similar stress-strain response to that of cast alloy and gold alloy, all of which showed work-hardening effect. Titanium and amalgam had curves that showed lower stress at comparable strain than enamel and showed no work-hardening. Amalgam exhibited the greatest creep behaviour, followed by Titanium, enamel and gold alloy. Similar to HAP, the cast alloy had very limited creep response. Only enamel showed significant backcreep at minimum load. This may be a consequence of the memory behaviour of the minor protein component within enamel.

Conclusions: The small remnant volume fraction of protein fragments have endowed enamel with metallic-like mechanical properties, which impart it with an ability to sustain repetitive cyclic contact loading over the life of the host. It may be better to choose metallic-like dental restorative materials to sustain the severe cyclic contact behaviour experienced and to protect the opposing teeth from excessive damages.

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1. Introduction

Enamel is generally considered a natural biological ceramic material¹ because of the “brittle” behaviour of its primary component, namely hydroxyapatite. During the past several decades, the development and designation of dental materials has been partially influenced by this understanding. As a result, more and more ceramic materials have been introduced into the dental clinic as restorative and teeth replacement materials. Of course, other considerations such as aesthetic requirements, biocompatibility and mechanical

properties have influenced selection. For the patients and commercial dental restorative suppliers, aesthetic effects and toughness of their products have recently come to dominate materials selection; however, this neglects the possible contact interactions between materials and opposing/adjacent teeth. From a materials engineering point of view, to avoid excessive damages of any part of the contact pair, it is better to choose materials with similar mechanical responses. From a dental clinical point of view, proper occlusal function has greater importance than aesthetic considerations for all dental restorative materials.

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As the antagonistic counterpart of most prostheses, enamel consists of 94–96 wt.% of hydroxyapatite crystals,² with the remainder mainly organic components and water. During the maturation of enamel, the protein matrix regulates mineralization, controls the morphology of the crystallites and may act as a mineral nucleator.³ At the time of mineral deposition, proteinases are also active in selective protein degradation and removal.⁴ As a result, after maturation, most of the protein matrix has been removed and only small amount of protein fragments, approximately 3–5% by weight, remain within the highly mineralized enamel. These proteins are mainly fragments of acidic proteins, enamelines and tuftelins.⁵ The result of such a highly precise regulation of the bio-mineralization procedure is the prominent hierarchy of enamel structure.^{6–8} The smallest structural units are needle or plate-like hydroxyapatite crystallites which are ~20 nm in thickness. In healthy human enamel, these hydroxyapatite crystallites are organized and bundled together by organic molecules into larger-scale structures, the so-called 'keyhole-shaped' enamel prisms. At this level, the arrangement of hydroxyapatite crystallites varies in their directions. Crystallite plates in the centre of prism are parallel to the prism while those near the edge of the prism usually have an angle of near 60° to the longitudinal axis of the prisms.⁹ The arrangement of prisms determines the next level unit, enamel types. Human enamel has the type which consists of ~5 µm diameter prisms encapsulated by thin protein rich sheathes that are arranged parallel in a direction perpendicular to the dentino-enamel junction (DEJ) from dentin to the outer enamel surface. In some areas, prisms may twist together or change their direction slightly to reinforce the whole structure.^{10,11} These compositional and structural characters endows enamel special properties such as anisotropic elastic modulus,¹² effective viscoelastic properties,¹³ much higher fracture toughness⁹ and stress-strain relationship more similar to metals than ceramics.¹⁴

By considering the above specialties of enamel, to better assist selection of dental materials, a number of commonly used dental metallic materials were selected to compare with enamel in more detail by nanoindentation method. The aim of the study is to verify that, although it has the basic composition of brittle hydroxyapatite crystallites, enamel behaves mechanically more as a metallic-like biocomposite.

2. Experimental methods

A Berkovich and a spherical indenter with a nominal radius of 10 µm (Synton, Switzerland) were selected for this study. Before the test, a standard fused silica specimen was chosen to calibrate the indenters.

Three healthy premolars extracted for orthodontic reasons were selected, disinfected with Milton solution and stored in distilled water till sample preparation. Several commonly used dental metallic materials were chosen for comparison with sound tooth enamel extracted for orthodontic reasons. Pure hydroxyapatite (HAP) was selected as a ceramic equivalent of enamel. The metal materials are cast alloy (Wiron® 99, BEGO, Germany), type IV gold alloy (ITI® Straumann, Switzerland), commercial pure Titanium (T-alloy, GC, Japan) and

amalgam. The amalgam sample came from a tooth with a class I amalgam filling extracted because of periodontal disease. Optical microscopy observation on the mesial-distal section of the sample illustrated that the filling was clinically successful.

All samples were cut, embedded with a cold-curing epoxy resin (Epofix, Struers, Copenhagen, Denmark) and polished (RotoPol-22, Struers, Copenhagen, Denmark) gradually from 230# sandpaper to 1 µm diamond polishing paste. After cutting and polishing, the inner part of whole enamel layer was exposed and the direction of enamel was selected so that the loading direction was parallel to the long axis of the enamel prisms. Once the tooth had been prepared, it was stored in distilled water at room temperature. All the other samples were stored at room temperature in dry laboratory conditions.

The indentation experiments were performed using a nano-based indentation system (Ultra Micro-Indentation System, UMIS-2000, CSIRO, Australia). The finished specimens were mounted on metal bases and held on the machine with magnetic force. All the tests of ceramics and alloys were done under dry conditions while enamel was tested in distilled water. Based on previous investigations by Poolthong,¹⁵ at loads comparable with those used for this study, there was no discernable change in the long-term stability of the hardness and elastic modulus of enamel stored at 4 °C in de-ionized water.

Elastic modulus and hardness of specimens were measured with the calibrated Berkovich indenter. The force applied during testing ranged from 10 to 300 mN, and was dependent upon the evaluated material. The software calculated elastic modulus and hardness as a function of penetration depth, h_p , for each indentation. At least 15 indents at different forces were done on each material to acquire the average values.

All the samples were indented with the spherical indenter at loading forces from 1 to 400 mN to measure the stress-strain relationship. The H - a / R curves were generated using the methods proposed in our previous work.¹⁴

The Berkovich indenter was used to investigate the indentation creep behaviour of different materials at a force of 250 mN. One-step loading and unloading method was used with a maximum-force (250 mN) hold time of 900 s and upon unloading a minimum-force (5 mN) hold time of 900 s for enamel and amalgam and 300 s for other materials. All tests commenced after at least 2 h thermal soak period in the instrument, and thermal drift was deemed to be almost negligible (<0.05 nm/s) before each experiment.

For all the tests, the distance between indents was more than 40 µm to avoid the influence of residual stresses from adjacent impressions.

After indentation testing, scanning electron microscopy (SEM) (XL-30, Philips, Netherlands) was used to examine the residual indent impressions. Enamel specimen was etched with a mild polyacrylic acid (10%) solution (Dentin Conditioner Liquid, GC, Japan) to better highlight the structures and indentation damage. All samples were ultrasonic cleaned with pure ethanol for 10 s before gold coating prior to SEM. Amalgam was also investigated with energy dispersive X-ray analysis (EDX) to determine its composition using an area

collection method at 1250 magnification. EDX composition analysis was repeated five times at different areas and the average and standard deviation were calculated and reported in the next section.

3. Results

3.1. Indentation elastic modulus and hardness

The indentation elastic modulus and hardness measured are listed in Table 1. Here, the indentation hardness value is the contact pressure of the indenter on the surface of the sample. Indentation elastic modulus, E_{eff} , has the following relationship with the true elastic modulus of the material, namely:

$$\frac{1}{E_{\text{eff}}} = \frac{1 - \nu^2}{E} + \frac{1 - \nu'^2}{E'},$$

E , E' and ν , ν' are Young's modulus and Poisson's ratio of specimen and diamond indenter. We assume E' and ν' as 1050 GPa and 0.07, respectively. The values of our results are within the normal range of other reports.^{16–19}

From the results, it is evident that, with the exception of the cast alloy, most of other materials have similar elastic modulus values to that of sound enamel. In the case of the hardness, pure hydroxyapatite has a much higher hardness value than enamel, while other metallic materials have similar hardness values as enamel.

3.2. Stress–strain relationships

Based on Tabor's empirical relationship²⁰ and the more recent work by Lawn and co-workers,^{21,22} H – a/R curves can be used to represent the stress–strain relationship of materials. In Fig. 1, the elastic part of the materials (short dots lines) were calculated with Hertz equation:

$$H = \left(\frac{4E_{\text{eff}}}{3\pi} \right) \frac{a}{R},$$

H is the mean contact stress (hardness), a the contact radius and R is radius of spherical indenter and in which E_{eff} of each material was read from Table 1. The numerical predictions of elastic responses of materials coincided very well with the real spherical indentation data, especially for HAP. Fig. 1 shows that, although its major component is hydroxyapatite, enamel displayed similar stress–strain responses to that of the metallic materials, especially cast alloy and gold alloy rather than

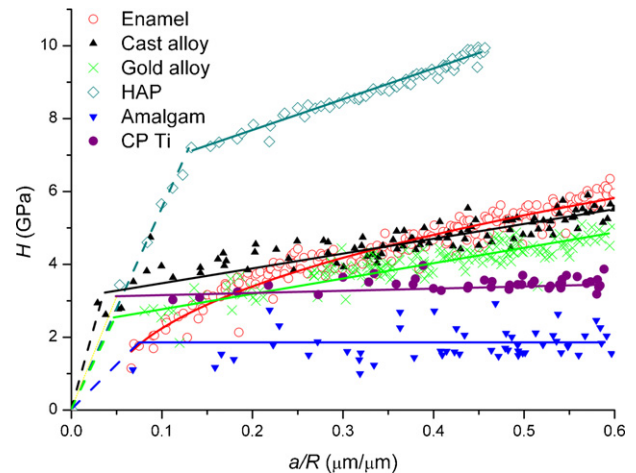


Fig. 1 – Stress–strain (H – a/R) relationship of different materials. Solid lines represent plastic part and short dots lines represent elastic part of materials, which was calculated from Hertz equation.

HAP, which had a well defined yield point at ~ 7 GPa. Enamel showed a smooth stress–strain curve without any prominent elastic–plastic transition point. It was not possible to predict the initial elastic response of enamel with Hertz equation

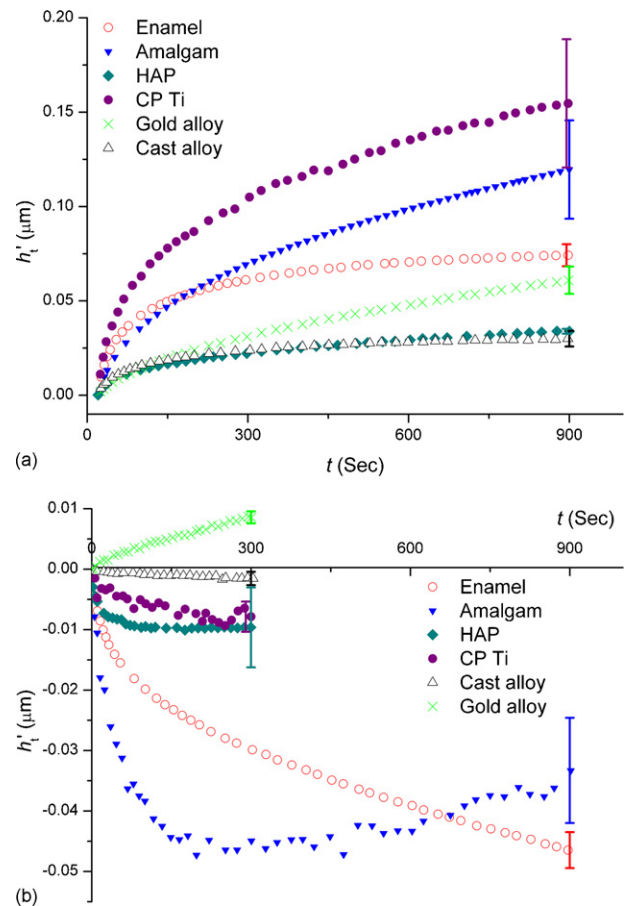


Fig. 2 – Indentation depth against time, h_t – t , curves of (a) creep and (b) backcreep of different materials (h_t values of amalgam in (a) were divided by 10 to fit onto the diagram).

Table 1 – Indentation elastic modulus and hardness of different materials

Material	Elastic modulus, E_{eff} (GPa)	Hardness, H (GPa)
Enamel	94.1 ± 5.4	4.2 ± 0.2
Hydroxyapatite	130.0 ± 4.7	8.1 ± 0.1
Amalgam	57.6 ± 9.2	2.1 ± 0.5
Cast alloy	209.8 ± 9.8	3.2 ± 0.1
Gold alloy	122.3 ± 4.5	3.5 ± 0.1
Titanium	145.1 ± 2.9	2.8 ± 0.2

because, in our preliminary tests, even with very small forces with a larger radius spherical indenter, no pure elastic responses were found. Also, HAP had much higher stress-strain curve than enamel. The flat horizontal stress-strain response beyond yielding for titanium and amalgam indicate that these two materials did not exhibit work-hardening.

3.3. Indentation creep behaviour

All the curves in Fig. 2 were the average value of at least 3 tests and the error bars at the end of the curves indicating the deviation of the data. The relative penetration depth, h'_t , was calculated by subtracting the initial depth at the beginning of the holding time from the increasing depths during the holding period at maximum and minimum loads.

Creep data at maximum-load holding exhibited increasing penetration depth with time for most investigated materials (Fig. 2a). Amalgam showed the maximum creep, followed by titanium. But they all exhibited considerable scatter in data. Enamel, on the other hand, had a relatively stable creep response. In contrast, the indentation creep depths of hydroxyapatite and cast alloy were very limited (less than $0.025\text{ }\mu\text{m}$ during the 900 s holding period). Such limited creep response, $\sim 0.03\text{ nm/s}$, is within the limit of UMIS system thermal drift stability and implied that the viscous properties of these materials are negligible compared with their elastic/plastic properties.

At the minimum-load holding period (named “backcreep” period), the recovery was prominent only for enamel and amalgam samples but remained within the instruments stability range ($<0.05\text{ nm/s}$) for other samples (Fig. 2b). The displacement rate and limited backcreep of hydroxyapatite ($\sim 0.03\text{ nm/s}$) and cast alloy ($\sim 0.01\text{ nm/s}$) were all within the limit of system thermal drift. Therefore, only enamel and amalgam exhibited significant backcreep response. When considered in more detail, the backcreep curves of enamel and

amalgam were totally different. Enamel showed a progressive recovery while amalgam recovered very quickly in the first 150 s and reached the limit of nearly $-0.05\text{ }\mu\text{m}$ in h'_t values. The latter portion of the amalgam curve only reflects system thermal drift.

This part of results implies that enamel and some dental metallic materials all exhibit creep behaviour but with different characteristics. This indicates different mechanisms are responsible for this behaviour and will be discussed in detail in the discussion.

3.4. SEM

Fig. 3 compares the residual indentation impressions and surrounding deformation for different materials. HAP exhibited typical brittle behaviour under sharp indentation. At loads of 200 mN, radial and lateral cracks formed with the latter extending to the surface resulting in chipping of the sample (Fig. 3b). In contrast, enamel showed residual deformation without any prominent cracks (Fig. 3a). The white arrow in Fig. 3a illustrates that even at a prism gap at the boundary of the indent impression, no crack formation occurred. Because of its low hardness, amalgam had larger indents at lower forces (Fig. 3c and d). Also, as is typical with plastic metallic material, no cracks were detected around the indents in amalgam. Comparison of the residual impressions for creep tests with loading-unloading test, indicated that creep tests induced some additional plastic deformation along the edge of indent (black arrow in Fig. 3d). Because of the morphological similarities, indentation impressions on other metallic materials were not shown.

As shown in Fig. 4, EDX spectra of amalgam indicated that, the main elements are Hg, Ag and Sn, with just little Cu. Element analysis indicated the composition of the sample is Ag $42.48 \pm 0.69\%$, Hg $33.89 \pm 0.7\%$, Sn $21.74 \pm 0.62\%$ and Cu $1.9 \pm 0.09\%$. This product may be considered a low copper

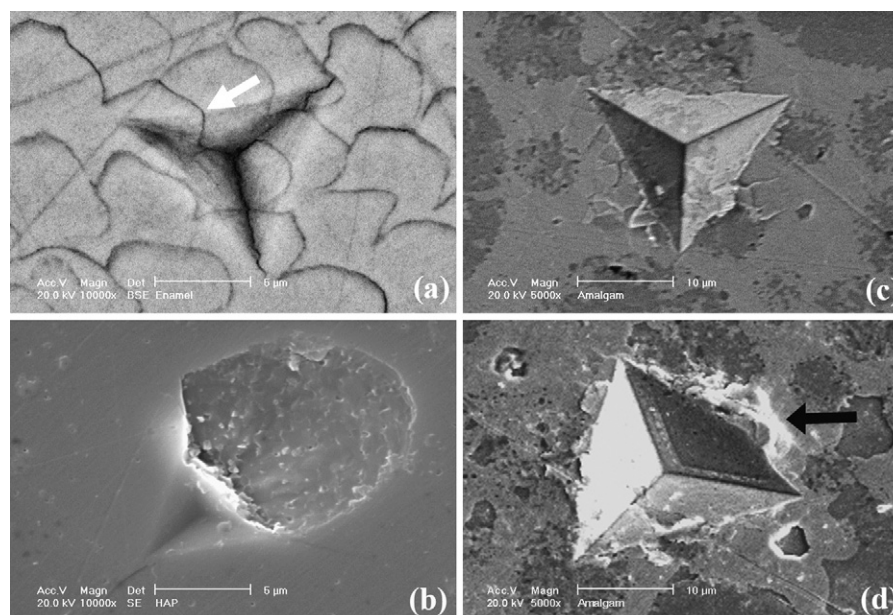


Fig. 3 – SEM pictures of Berkovich indenter impressions on different materials: (a) enamel after 250 mN loading, (b) HAP after 200 mN loading, (c) and (d) amalgam after 150 mN loading and 250 mN plus 900 s holding, respectively.

Label A: Amalgam

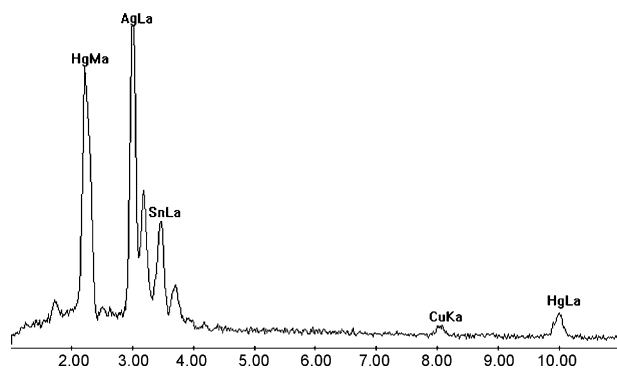


Fig. 4 – Sample EDX spectra of amalgam sample showing the major Hg, Ag and Sn peaks with minor amounts of Cu.

amalgam. The dark area of amalgam contained mainly Ag and Sn while the bright area only contained Hg and Ag.

4. Discussion

4.1. Differences between enamel and HAP

HAP, a typical crystalline ceramic material, is composed of sintered small apatite crystals with minimal dislocations present within crystal lattices and at the boundary of adjacent crystals. This predominant ceramic character determines the mechanical properties of HAP as a hard and brittle material. The indentation stress–strain behaviour of HAP was similar to that previously reported for a range of porcelain and restorative ceramic materials.¹⁴

Although the primary component of enamel is also hydroxyapatite, the mechanical responses of enamel were totally different to those of pure HAP, such as lower elastic modulus and hardness (Table 1), lower stress–strain response (Fig. 1), and no brittle cracks about indentation impressions (Fig. 3). These results confirm the declaration that the hierarchy of enamel structure and the small protein fragments within the enamel may regulate the mechanical properties of enamel, as mentioned in Section 1.

Protein films act as the cushion between the crystallites and prisms. The inelastic properties of proteins like biopolymers can help to dissipate and redistribute the stresses and protect crystallites from fracture.²³ Structure hierarchy also regulates the mechanical properties of enamel at the prism level. Nano-scale investigations^{24,25} have shown that, compared to those in the head area of the rods, elastic moduli and hardness were lower in the tail area and in the inter-prism enamel, which can be attributed to changes in crystal orientation and the higher content of soft organic tissue in these areas. Moreover, the staggered mineral crystallites may deflect and bridge the cracks and strengthen the bulk material on a large scale. As a result, enamel has higher fracture toughness than pure hydroxyapatite.⁹ This has also been confirmed by our SEM pictures which showed that even the sheath area at the contact boundary, the presumed weak area, did not form any cracks (white arrow in Fig. 3a).

4.2. The metallic-like deformation properties of enamel

From a functional point of view, enamel needs metallic-like deformation properties to withstand millions of cyclic occlusal loadings. A comparable example to understand this requirement is, for some mechanical structures needing to sustain cyclic loadings such as bridge frames, we prefer tough ductile materials rather than stiff but brittle ceramics. The reason is metals are more plastic deformable and flaw insensitive and are better able to withstand random overloads as cyclic-force bearing structures. During its evolutionary development, enamel has been optimized structurally and endowed with metallic-like deformation properties to sustain multiple impact and sliding contact events.

In the field of materials evaluation, one needs different parameters to rank materials for different aspects. Elastic modulus or the ratio of stress to strain below the proportional limit and can reflect the elastic deformation ability of a material; the hardness of a material is usually considered as the resistance to permanent indentation deformation; while the stress–strain response reflects the elastic–plastic mechanical behaviour of material usually on a large scale and creep behaviour is a reflection of viscous property of a material. These basic parameters reflect the mechanical properties of materials and, therefore, may be used to compare and select materials for specific applications. In the present instance, we wish to compare enamel with the other materials we have investigated.

Firstly, enamel has similar elastic modulus, hardness and stress–strain responses to those of widely accepted dental metallic materials. By comparing with hard brittle materials such as ceramics, lower hardness values of metals mean these materials have better plastic deformation abilities. Plastic metallic materials can dissipate more energy before catastrophic failure. SEM comparative observations also illustrated that elastic–plastic materials such as enamel, gold alloy and amalgam could survive under bigger loading forces without cracking. Moreover, enamel showed very similar stress–strain response to those of metals rather than ceramics. From these perspectives, metallic materials may be more suitable than ceramics for dental fillings and crown/bridge because of similar basic mechanical responses.

Previous studies have shown that gold alloy causes less wear on the opposing dentition than dental ceramics.^{26–28} This observation is supported by our previous investigation of the stress–strain relationship of a number of dental ceramic materials,¹⁴ where it was shown that the onset of non-linear behaviour occurred at much higher stress levels than enamel. It is thus apparent that the stress–strain properties, which reflect the hardness, have a close relationship with the wear behaviour of materials and materials with similar stress–strain response as enamel may have less abrasive or better protective effects on opposing teeth. The detailed relationship between stress–strain response and wear property of materials will be an interesting direction of future studies.

Secondly, the indentation creep ability of enamel was much greater than HAP but similar to some dental alloys. A recent literature review of Li et al.²⁹ showed that most materials, including hard ceramics, have limited ability to creep at room temperature. The theoretical analysis of

indentation hardness test by Johnson³⁰ suggested when an indenter is pressed into a flat surface of a homogeneous material, the indenter is encased in a non-deforming hemispherical “core” of material, within which there is assumed to be a hydrostatic pressure. Surrounding this hydrostatic zone, there is a plastic–elastic hemispherical zone. The process of indentation is thus likened to the plastic movement of a series of shells concentric with the hemispherical core. Li et al.²⁹ also summarized the major possible mechanisms of indentation creep for different engineering materials as: plasticity (dislocation glide), power law creep (climb plus glide), power law breakdown (glide plus climb), recovery (dislocation climb), and diffusion (volume, grain boundary or dislocation pipe). Reports on a titanium alloy described the mechanisms of creep as the rearrangement of the atoms within crystal lattice, along with formation, movement and pile-up of the dislocations, slip of crystals along grain boundaries and so on.^{31,32} All of these mechanisms play a role in the expansion of the elastic/plastic boundary into the material under the contact area at maximum load, and the speed of this expanding procedure determines the creep ability of the material.

Here, because of the structural complexity and similarity in creep mechanisms of other alloys, we consider only amalgam as an example to illustrate the mechanisms of indentation creep on metallic materials. Generally speaking, after setting, an inter-metallic compound of silver and mercury, Ag_2Hg_3 (γ_1 -phase), and a solid solution of mercury in tin, Sn_7Hg (γ_2 -phase), form the matrix of the amalgam (bright area in Fig. 3c and d), and some un-reacted Ag_3Sn immersed in the matrix (dark area in Fig. 3c and d). Moreover, some pores and fissures may be generated at the boundary interface of different phases during the setting procedure (pores in Fig. 3d are good examples). These defects and the multiplicity of phases influence the mechanical properties of the material dramatically. This can also partially explain the large data scatter of the material in all the amalgam tests of this study. Theoretically, creep in amalgam occurs by plastic flow and grain-boundary sliding.³³ According to Xu et al.,³⁴ subsurface creep of indentation area is dominated by deformation of individual alloy particles as well as matrix grains, which are associated with the creep-induced shape changes of individual alloy particles from spherical to irregularly elongated shapes, and the matrix grains separating them from each other possible due to plastic flow and/or grain rotation. The black arrow in Fig. 3(d) also pointed out possible creep damage at the edge of an indent, along with some additional plastic deformation.

Because of its totally different chemical composition, enamel will have other creep mechanisms than occur in metals, mainly related to the organic protein components existing between the apatite crystallites. This proposition is supported by research on creep of bone,³⁵ in which the stress relaxation curves of bone were found to be similar to those of demineralized bone. Moreover, in comparison with HAP sample, enamel exhibited a much more extensive creep response. All of these observations indicate that proteins were the dominant contribution to creep behaviour. Therefore, it is safe to conclude that creep behaviour of enamel comes mainly from the protein films between the apatite crystallites and the prisms. Moreover, the limited creep response may be because

the protein films occupy only a very small volume fraction of the whole enamel.

Most proteins may be regarded as bio-polymers. Like normal polymeric materials, they have viscous properties. Moreover, proteins inside enamel may also contain special bio-molecular structures such as “sacrificial bonds”.^{36,37} This type of bond has saw-tooth shaped force–displacement curves because of the gradual break down of the domains and time dependent unfolding of the peptide chains. This may help us to explain the observed creep of enamel. Another characteristic of this bond is, after unloading, the bond can recover and the peptide chain can be refolded. The reversibility of this bonding coupled with the residual stress surrounding the deformed zone about the indentation impression may explain the prominent backcreep observed for enamel.

Thirdly, only enamel showed stable backcreep upon partial unloading. Backcreep may be regarded as the behaviour of polymers, which have the ability to reform visco-elastically after unloading. For HAP and most metallic materials, plastic deformation, no matter if it is viscous or not, is permanent and unrecoverable. This also suggests that the more deformable organic protein-rich material in the sheath region and between the apatite crystallites assisted the recovery of the deformed material back towards its initial position. This property may be important in enamel’s ability to function as a long-time force bearing tissue and contribute to the superb anti-fatigue behaviour of enamel.

The mechanism of initial rapid back-creep of amalgam is not clear but may be a consequence of similar creep mechanisms as for enamel producing a visco-elastic recovery of the deformed material under the indentation residual stresses. In comparison with the extensive creep of amalgam at peak load, the extent of the backcreep was not significant (only ~5%) whereas in the case of enamel backcreep was much more significant (~60%).

5. Conclusion

Comparison between the indentation responses of enamel and several commonly used dental metals showed that enamel, although the major component is brittle hydroxyapatite, behaved more metallic like with respect to its stress-strain curves, creep and crack initiation behaviour. This investigation indicates that dental metallic alloys may have better mechanical compatibility than ceramics with respect to enamel. Moreover, rather than depending on dislocations as metallic materials, enamel acquires its metallic-like properties as a consequence of the limited presence of proteins within it. This may provide materials researchers with an alternative novel micro-structural design for dental restorative materials.

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